iv. Edaphic Stress

They are divided into two types. They are **water stress** and **salt stress:**

a. Water stress

A common stress condition arising from lack of water or excess of water is called **water stress**. The abundance of water leads to a stress called **flood stress** and scarcity of water leads to a stress called **drought stress**.

I. Flood Stress

The temporary inundation of plants and its parts by flooding causes oxygen deficiency to the roots and soil borne microorganisms. Effects of flooding are as follows: Nitrogen turnover in the soil is reduced; Abscisic acid, ethylene and ethylene precursors are formed in larger amount; Stimulation of partial stomatal closure, epinasty and abscission in leaves; Cellular membrane systems break down, mitochondria and microbodies disintegrate and enzymes are partially inhibited. Flood tolerant plants include those found on permanently wet soils. Examples: Marsh plants, shore plants and hydrophytes. Tree species found dominant in flooded sites are also tolerant. Examples: *Taxodium disticum,* Mangroves and palms are tolerant to flood stress.

II. Drought Stress

The term 'drought' denotes a period without appreciable precipitation, during which the water content of the soil is reduced to such an extent that plants suffer from water deficiency. Effects of drought are as follows: Decrease in cellular growth and synthesis of cell wall components cause the cells to become smaller in size; Nitrogen fixation and its reduction are decreased by decreasing the activity of certain enzymes; Increase in abscisic acid level ultimately closes down the stomatal apparatus to the minimum, hence, transpiration declines; Protochlorophyll formation is inhibited and photosynthetic process declines; ; Wilting in mature leaves is associated with carbohydrate depletion due to mobilization export, followed by leaf senescence.

Mechanism of drought resistance

Xerophytes are well adapted for drought either because,

i. the protoplasm of such plants does not die when it faces extreme or prolonged desiccation (dehydration) hence, it tolerates or endures such conditions. Example: Creosote bush (*Larrea tridentata*) can survive water content drops upto 30% whereas, in most plants the lethal level is below 50–70% or these plants are able to avoid or postpone the lethal level of desiccation because they have developed structural or physiological adaptations. Plants have developed following mechanisms:

- To avoid or postpone desiccation improved water uptake by developing the roots to deep water source
- Improved water uptake by roots which penetrate deep water source;
  Increasing and enlarging the conductive tissues.
- Stomata present only on the lower epidermis with by dense trichomes;
- Rolling of leaves and water storage in succulent tissue.

During drought stress an essential protection mechanism that stabilizes the cell structure is induced gene expression of **stress protein (dehydrin and osmotin)**. These proteins protect the macromolecules in the cytoplasm and in the nucleus, the cytoskeleton (biomembranes) against denaturation. High desiccation tolerance implies that the protoplasm rehydrates when water becomes available. Plants growing in deserts and arid regions are usually drought resistant.



Resurrection plants, those plants having ability to survive near total drying which causes them to appear dead. They recover when water is available. Example: *Selaginella lepidophylla*

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